

Capstone Project

Marketing Campaign Customer Segmentation

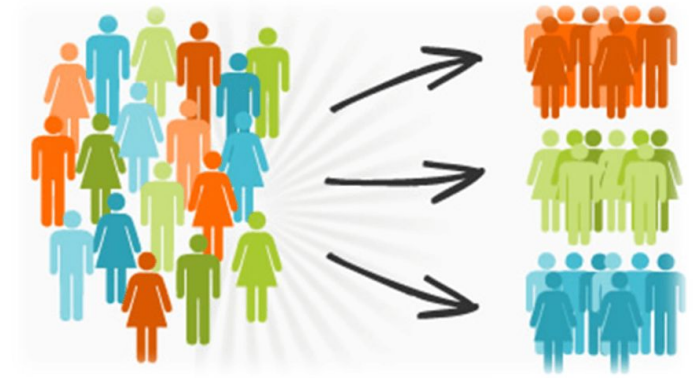
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Final Report Presentation
17th December 2022

Executive Summary

Unsupervised Learning - **Clustering**

To gain insight into customers:

- Personalized marketing strategy



💎😄 Who are the most valuable customers? 💎😄

😄💰 Who are the highest potential customers? 😄💰

🤝👍 Who are most responsive customers to marketing campaigns? 🤝👍

😞👎 Who are the 'low value' customers? 😞👎

Executive Summary

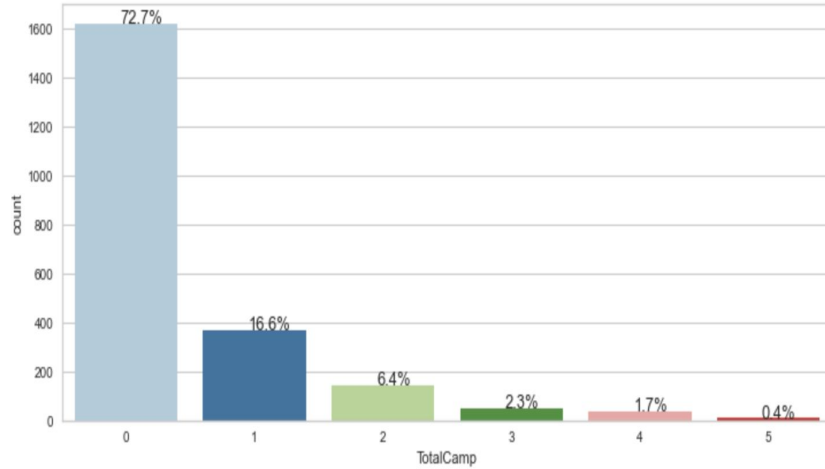
This project proposes the adoption of **K-Medoids Algorithm Solution with four clusters** into the following personas:

- **High Value**
- **High Potential**
- **High Loyalty**
- **Low Value**

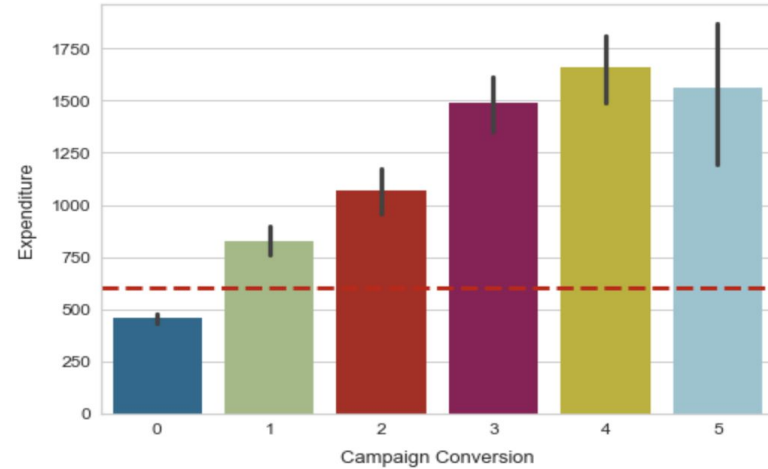
With these four customer profiles, the following are **strategic implementation recommendations**:

- **Understand** the unique needs of each profile
- **Prioritize** target customer profiles
- **Align** customer strategy with the rest of company
- **Assign** KPIs - **test, iterate and optimize**
- **Run** personalized marketing campaigns

Problem Statement



Low campaign conversion.



Higher customer monetary value

The goal is to use customer segmentation to improve campaign conversion and thus customer monetary value

Solution Roadmap



Exploratory Data Analysis

The dataset contains **three years of customer records from a retail grocery business**. The original dataset contains 27 features belonging to over 2,200 customers.

The dataset consists of both numerical and categorical data, and can be grouped into:

- Customer demographics
- Customer spend per product category
- 6 promotional campaigns' conversion rates
- Retail channels

Feature Engineering & Feature Selection

Then, the data is preprocessed before it is inputted into the clustering algorithms.

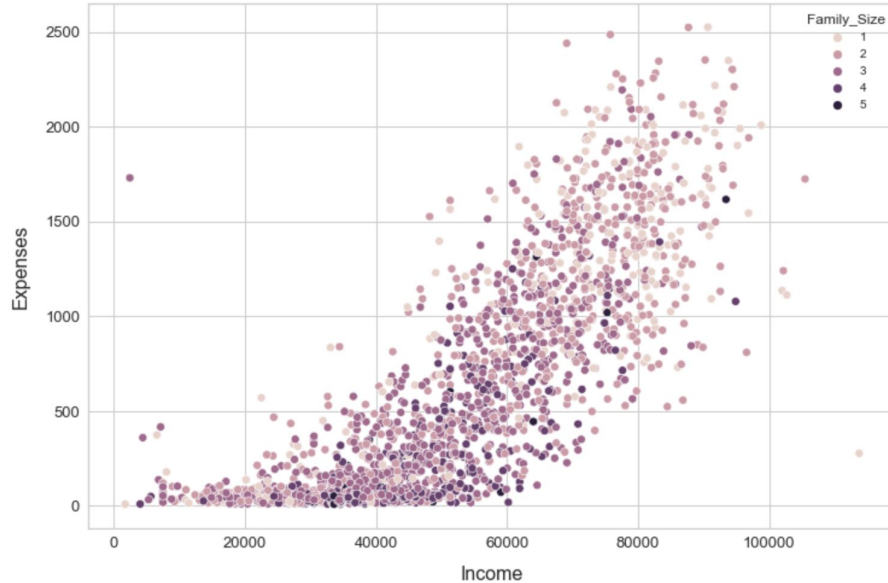
Feature engineering and data preprocessing include:

- Removing unnecessary features, outliers and imputing missing values
- Recategorizing features (such as family size, relationship status, product categories)
- Transforming and creating new features that give better insight into the analysis (age, summing up related categories)

After this step, we use **15 features** for clustering analysis.

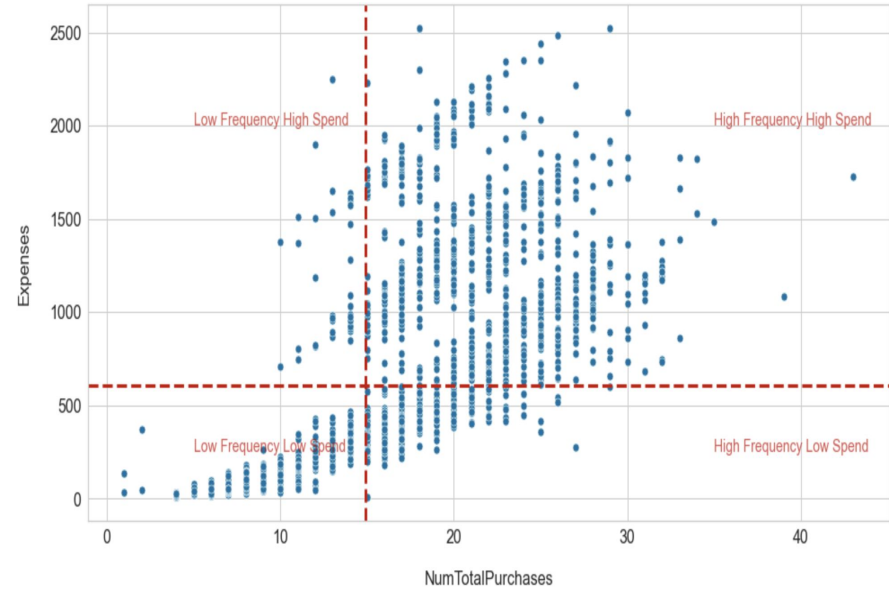
Insights from the data

**Purchasing Power:
Income vs Expenses**



The plot shows a positive correlation between Income and Expenditure at the store, with lower family sizes associated with higher income and expenditure.

**Purchasing Behaviour:
No. of Purchases vs Expenses**



The plot above shows the scatterplot of Number of Total Purchases vs Expenses (total expenditure)

Evaluation of Clustering Models

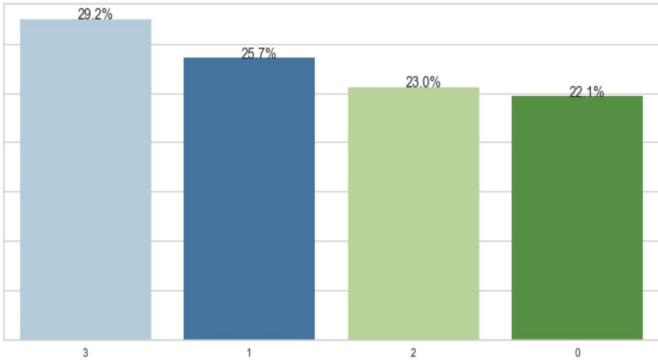
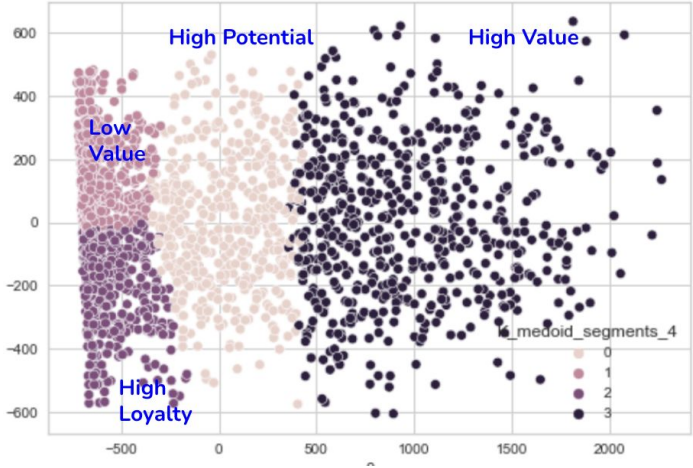
Algorithm	# of k	Silhouette Score	Properties of clusters
K-Means	4	0.382	Distinct and unique customer profile
K-Medoids	4	0.341	Distinct and unique. Customer profile with largest monetary value
Hierarchical Clustering	6	0.298	Overlapping clusters for outliers. Not distinct. Separates high value customers into 3 separate clusters
DBSCAN	3	0.358	2 large clusters with 1 with small outliers. Not suitable for this dataset
GMM	3	0.289	Undistinct and overlapping clusters. Customer profile not insightful.

K-Medoid produces the most distinct and unique customer profiles for personalized marketing strategy.

k=4 returned the highest silhouette score for K-Medoids (see appendix)

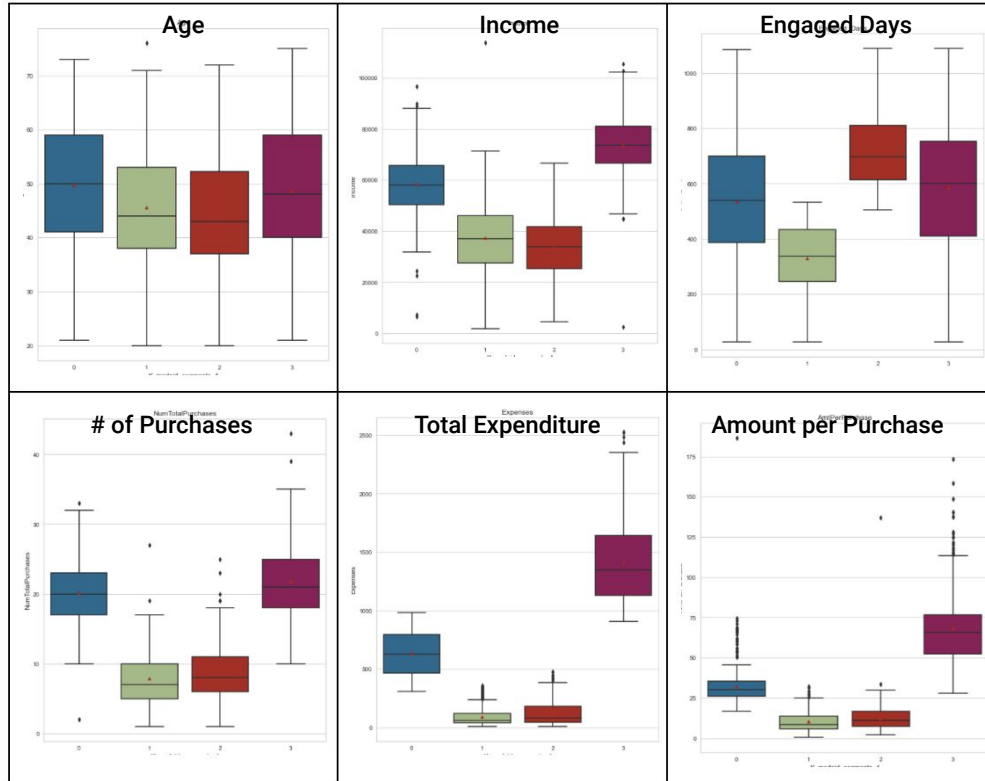
Final Solution Proposal: K-Medoids, k = 4

Cluster	% of customer	Customer Persona	Customer Characteristic	Action
0	22%	High Potential	High frequency and deal-sensitive customers. Buyers across all channels, esp online	Show upgraded product recommendations to increase basket size Identify categories that are ignored.
1	26%	Low Value	Low-spending newest customers	Survey for better feedback and improvement
2	23%	High Loyalty	Oldest customers	Loyalty discounts for higher purchase frequency
3	29%	High Value	High income. Big spenders.	Reward with VIP tiering with premium service. Reduce churn.



High Value customers as the largest percentage of the dataset

K-Medoids Customer Profile



K-Medoids gives us the highest ROI:

- **High Potential** - high frequency leads to upgrade / upsell potential
- **Low Value** - either ignore for cost-saving or potential for feedback to upsell
- **High Loyalty** - Loyalty discounts
- **High Value** - Reward with VIP tiering premium service.

Recommendations and Next Steps

Initial Next Steps (next 3 months)

- **Understand** the unique needs of each profile (survey, A/B testing, conjoint analysis)
- **Prioritize** target customer profiles
- **Align** customer strategy with Product, Technology, Operations, Customer Service
- **Assign** KPIs and compare between cluster profiles - **test, iterate and optimize**
 - Conversion Rates, Monetary Value, Net Promoter Score
- **Run personalized marketing campaigns based on the above steps**

Additional Next Steps for Further Analysis (6 months+)

- Cross-sell between channels and recommend products based on **Clustering-based Recommendation Systems** (Collaborative Filtering) and **Network Analysis** on products and brands.
- Run **Decision Tree** algorithms against predicted clusters

Future Next Steps (1 year+)

- **Decision to rerun** clustering algorithms -
 - Quarterly or Annually weighed against Cost Benefit analysis

Limitations and Challenges

- **Data quality and suitability critical for analysis**
 - Future data collection should include customer preferences
- **Interpretation** of the resulting customer segmentation is crucial for developing a successful marketing strategy
 - Domain knowledge matters
- **Initial assumptions matter and affect results**
 - Feature selection, dataset, number of clusters
- **Frequency** at which analysis is rerun should be determined based on the company's needs

Appendix

Insights from the data

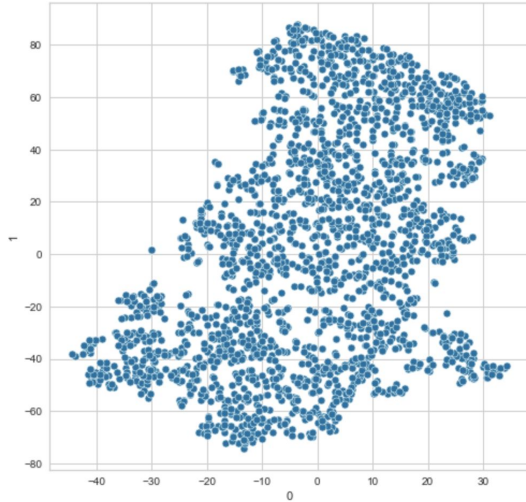
Correlation Matrix

NumStorePurchases	0.68	0.83	0.52	0.17
NumCatalogPurchases	0.80	0.74	0.68	0.37
NumWebPurchases	0.53	0.79	0.38	0.20
NumDealsPurchases	-0.08	0.35	-0.19	-0.09
NumWebVisitsMonth	-0.50	-0.32	-0.50	-0.13
Wine	0.90	0.72	0.82	0.49
FishMeat	0.87	0.58	0.84	0.33
Dessert	0.69	0.53	0.65	0.21
Gold	0.53	0.50	0.49	0.20
	Expenses	NumTotalPurchases	AmtPerPurchase	TotalCamp

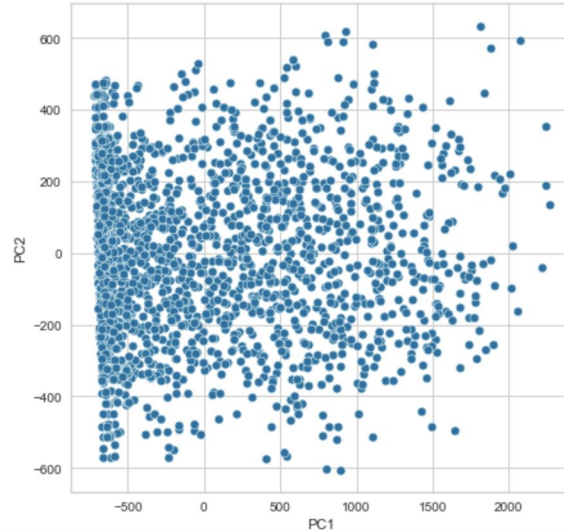
- We can see a **high positive correlation** between:
 - **Expenses** and **Catalog Purchases**, the highest amongst the channels.
 - Amongst products, **Expenses** has the highest correlation amongst **Wine** and **FishMeat**
 - **NumTotalPurchases** and **Wine** - being the highest, vs the other products in the 0.50s.
 - **NumTotalPurchases** and **Store, Web** and **Catalog** compared to **Expenses** which has the highest correlation with **Catalog** purchases amongst the different retail channels. This suggests that **Wine** sales are primarily done through the **Catalog** channel
- We can see a **negative correlation** between:
 - **WebVisits** and all the product categories, as well as **Catalog** and **Store** purchases. (though positively correlated with **DealPurchases** suggesting that customers tend to look online for discounts.

Dimensionality Reduction from 15 features to two (90% variance)

Stochastic Neighbor Embedding (t-SNE)



Principal Component Analysis (PCA)



Each principal component is a linear combination of the original features.

- In these results, the 1st principal component (PC1) has large positive associations with **Expenses** and **Wine**, and to a lesser extent, **FishMeat**. We can interpret this component to be primarily a measure of a customer's expenditure at the store, since we see the high correlation between these variables above in the correlation matrix.
- The 2nd principal component (PC2), where the data spreads the 2nd most, has a large negative association with **Engaged_Days** - how long a customer has been a purchasing from the company.

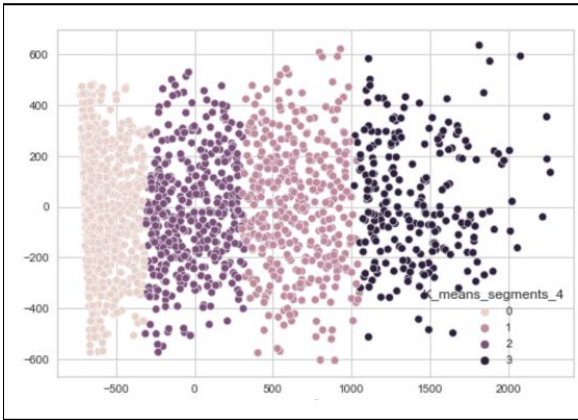
Plots of t-SNE and PCA show the dataset is not well separated with any clear pattern

Clustering Algorithms used on Dataset

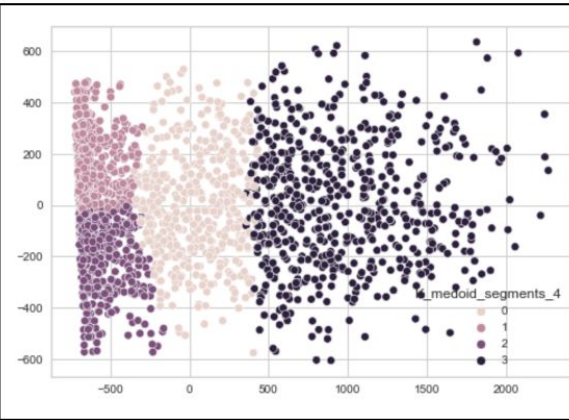
The following four types are the most widely used types of clustering models that are applied to the dataset:.

Model	Method	Algorithm
Centroid Model	Uses distance between a data point and the centroid of the cluster to group data. Iterative.	K-Means K-Medoid
Connectivity Model	Uses closeness of data points to decide clusters	Hierarchical Clustering
Density Model	Scans the data space and assigns clusters based on the density of datapoints	DBSCAN (Density-based Spatial Clustering of Applications with Noise)
Distribution Model	Segments data based on probability of belonging to the same distribution	Gaussian Mixture Model (GMM)

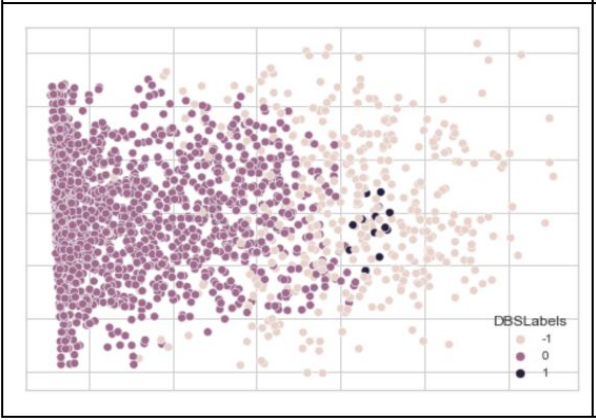
Comparison of Clustering results across algorithms



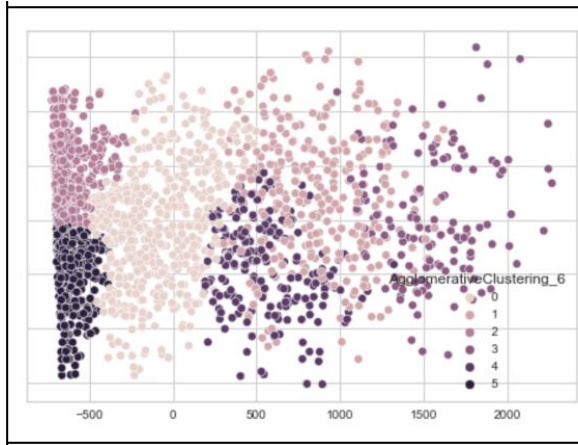
K-Means, k=4, SS= 0.382



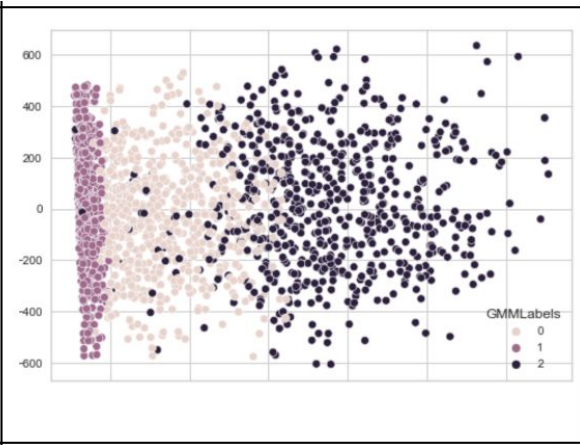
K-Medoids, k=4, SS= 0.341



DBSCAN, k=3, SS= 0.358



Hierarchical Clustering, k=6, SS= 0.298



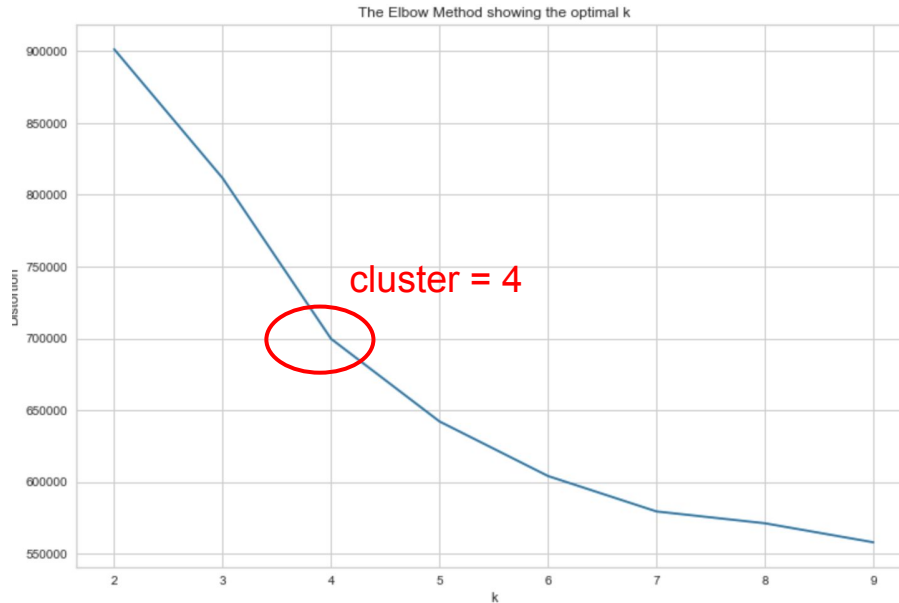
GMM, k=3, SS= 0.289

K-Medoids Customer Profiles

Customer Profile	
High Value	* High income, high frequency shoppers * Prefer in-store and catalog purchases * Highest purchase across all product categories, esp wine * Lowest number of web visits amongst customer profiles * High willingness-to-pay for premium service
High Potential	* Above average income * High frequency shoppers * Responsive to discount deals * Buyers across all channels, esp online * Oldest age-group customers
High Loyalty	* Lowest store purchase * Longest customer group (700+ days) * Highest Web Visits
Low Value	* Low spending customers * Newest customers

K-Medoids: Cluster Selection using Elbow Method and Silhouette Score

For `n_clusters = 3`, the silhouette score is 0.3362711791391004)
For `n_clusters = 4`, the silhouette score is 0.34101964644255905)
For `n_clusters = 5`, the silhouette score is 0.33104249174322775)
For `n_clusters = 6`, the silhouette score is 0.32894336694790344)



4 is the optimal cluster value based on:

- 1) Elbow method - gradient change (red circle)
- 2) Silhouette Score - highest where `n_clusters = 4`